STRUCTURE-FIRST-ML

Collected Technical Reports

Structure-first machine learning for physical data

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Preface. Most machine-learning pipelines coerce data into the shape the model expects: irregular observations are interpolated onto a regular grid, point clouds are voxelised, curved domains are flattened, and conserved quantities are left to be learned by accident. Each step discards structure that was present in the measurement and introduces artefacts that were not. The work collected here reverses that order of reasoning — identify the mathematical object the data belongs to, choose a representation with the invariances that object actually has, and only then attach a learning algorithm. Where the representation carries a theorem, the theorem is stated and its hypotheses are tested against real data rather than assumed. Results are reported in both directions: the cases where the approach wins and the cases where it loses are given equal space, because a representation that is claimed to help everywhere has not been understood anywhere.

Data courtesy of the Zwicky Transient Facility, the ZTF Bright Transient Survey, and the ALeRCE broker.

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Chapter 1

Path signatures for irregularly sampled light curves

projects/t1-signature-lightcurves

Abstract

A photometric light curve is a finite sample of a path, not a vector: observations arrive at times set by weather and scheduling, in whichever filter was in use, with per-point uncertainties varying by an order of magnitude. Standard pipelines destroy that structure at the first step by interpolating onto a regular grid. The signature transform of rough path theory consumes the raw stream directly and is invariant under reparameterisation of time. This chapter tests whether that theoretical advantage survives contact with real survey photometry, on 2,375 spectroscopically classified ZTF transients in eight classes. The answer is qualified, and the qualifications are the substance of the chapter.

1.1 Introduction

The signature of a path has been applied once in astronomy, to radio-frequency interference detection in interferometric visibility streams [1]. Time-domain photometric classification is untouched. That gap is narrow enough to be worth attempting and broad enough to be worth reporting either way.

The incumbent methods are not weak. Hand-crafted variability features in the lineage of FATS and `feets` have absorbed two decades of domain knowledge; Gaussian-process interpolation is the principled treatment of irregular sampling; random convolutional kernel methods such as MiniRocket [4] are cheap, deterministic and strong. A new representation earns its place only by beating these or by adding something they lack.

1.2 Mathematical background

Definition 1 (Signature)**.** *For a continuous path $X : [0, T] \to \mathbb{R}^d$ of bounded variation, the signature is the sequence of iterated integrals*

$$\mathbf{S}(X) \;=\; \left(1,\; \int_{0<t<T} \mathrm{d}X_t,\; \iint_{0<s<t<T} \mathrm{d}X_s \otimes \mathrm{d}X_t,\; \dots\right),$$

whose k-th term is an element of $(\mathbb{R}^d)^{\otimes k}$.

Three properties motivate its use here.

Reparameterisation invariance. For any increasing reparameterisation φ, $\mathbf{S}(X \circ \varphi) = \mathbf{S}(X)$. The signature encodes the order and shape of what happened, not the clock on which it happened. When observation times are set by the weather rather than by the source, this is the correct invariance to have; interpolation-based pipelines must learn it from data instead.

Faithfulness. The signature determines the path up to tree-like equivalence, informally up to retracing [2]. Nothing is discarded but the parameterisation.

Universality. Linear functionals of the signature approximate continuous functions on path space uniformly on compact sets. A linear model on signature features is therefore a universal approximator on paths, which converts architecture design into a question of truncation depth.

For a piecewise-linear path the signature has a closed form. Each straight segment with increment δ contributes the tensor exponential $\delta^{\otimes k}/k!$ at level k, and segments concatenate by Chen's identity

$$\mathbf{S}(X * Y)_k \;=\; \sum_{i=0}^{k} \mathbf{S}(X)_i \otimes \mathbf{S}(Y)_{k-i}.$$

Truncating at depth m retains $\sum_{k=1}^{m} d^k$ coefficients.

1.2.1 Implementation and verification

The transform is implemented from scratch rather than imported, because the theory track requires access to individual tensor levels and because the standard C++ backends do not build reliably on current Python versions. Correctness is established two ways: agreement with `esig`, `signax` and `roughpy` to $\sim 10^{-13}$ across four path configurations, and four structural identities checked directly — reparameterisation invariance, level one equalling the total increment, the shuffle identity $S_1 S_2 = S_{12} + S_{21}$, and the vanishing of the signature of a path concatenated with its own reversal. All sixteen checks pass.

1.3 Where the ordering information lives

Before applying the transform to real data it is worth confirming that the claimed mechanism exists, in a setting where the answer follows from the construction. Two classes of synthetic double-peaked light curve are built to differ only in which peak comes first: the peaks sit symmetrically about the midpoint of the window and the curve starts and ends at the same baseline, so one class is the exact time reverse of the other. The marginal distribution of magnitudes is therefore identical between classes, and so is the net change from first to last observation.

The prediction recorded in advance was that separation would appear at depth 2, the Lévy area being the classical order-sensitive term. It does not.

The reason is geometric and specific to this path construction. With channels (t, m) the time coordinate is strictly increasing, so the antisymmetric part of level 2 — the Lévy area —

Representation	Balanced accuracy
Order-blind summary features	0.512
Signature depth 1	0.524
Signature depth 2	0.489
Signature depth 3	**0.978**
Signature depth 4	0.990

Table 1.1: Separation of two synthetic classes differing only in the order of two peaks. Chance is 0.5. Ordering becomes visible at level 3, not level 2.

reduces to $\int m\,\mathrm{d}t$ once the boundary terms vanish. That is the area under the light curve, which is the same whichever peak came first. A path monotone in one coordinate encloses no signed area that ordering can change. Ordering first becomes visible at level 3, in terms of the form $\iiint \mathrm{d}t\,\mathrm{d}m\,\mathrm{d}t$, which weight magnitude excursions by when they occurred.

The practical consequence is direct: truncating at depth 2 to economise on features would discard precisely the information the method exists to supply, and the resulting null result would have been attributed to the astrophysics.

1.4 Data

Labels are the spectroscopic classifications of the ZTF Bright Transient Survey; per-epoch photometry in g and r is served by the ALeRCE broker. ALeRCE's own machine classifications are never used, since training on one classifier's output and reporting agreement with it measures nothing.

After quality cuts requiring at least ten detections and at least three in each band, the sample is 2,375 objects with a median of 42 detections spanning a median of 122 days, in eight classes: CV (456), SN II (426), SN Ia (406), AGN (376), SN IIn (273), SN Ic (185), SN Ib (148) and SLSN-I (105).

A first version of the fetch submitted objects grouped by class while the broker's failure rate rose during the run, so classes submitted last lost far more objects than those submitted first and SN Ib was eliminated entirely. That is a selection effect on the label rather than random attrition, and it would have corrupted every downstream number. After shuffling before any network call, lowering concurrency and adding exponential backoff, the per-class retrieval rate is 1.0 across all eight classes.

1.5 Method

1.5.1 Constructing the path

The difficult part is not the signature but deciding what the path is. Observations in different filters are not simultaneous, so no instant exists at which a two-band vector is defined, and any joint multi-channel construction embeds a choice. Three are implemented and compared rather than assumed: *per-band*, in which each filter is its own two-dimensional path and the signatures are concatenated, with no cross-band information and no interpolation; *interleaved*, in which all observations in time order form one path with a band indicator channel; and *joint forward-fill*, a common time axis with the last observed magnitude carried forward, which is piecewise-constant interpolation and is labelled as such.

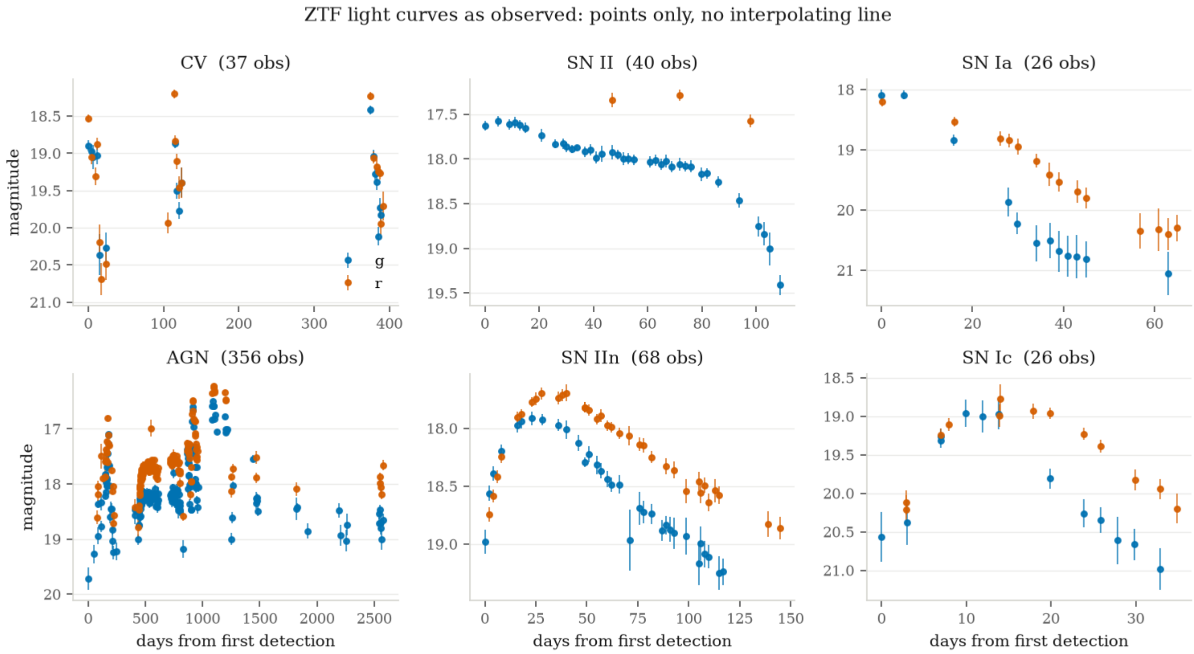

Figure 1.1: Example light curves, one per class, drawn as points with error bars and no connecting line: a line between observations would assert a continuity the data does not contain. Note the range of durations, from 26 detections over 65 days to 356 over 2,500.

1.5.2 Channel preparation

How the channels are prepared turned out to matter more than which path construction was used. Scaling time to $[0, 1]$ discards the duration of the event; centring magnitudes discards absolute brightness. Both are strong physical discriminants for transients, and the hand-crafted baseline retains them as `t_span` and `peak_mag`. Discarding them and then reporting that signatures lose would measure the preprocessing rather than the representation.

A related fact is worth stating because it is a check on the implementation rather than a result: the *raw* and *raw time* preparations score identically to four decimal places. The signature depends only on increments, so a constant magnitude offset cannot change it. Absolute brightness can enter only through a basepoint, and when introduced that way it degrades performance rather than improving it.

1.5.3 Baselines

Every representation is evaluated on identical cross-validation folds with identical classifiers. The hand-crafted baseline implements the FATS and `feets` lineage from scratch; almost all of its features are functions of the marginal magnitude distribution and are therefore invariant to permuting observations in time, which is exactly the information signatures retain. The three ordering-sensitive features it does contain — linear trend, rise and fade fractions, maximum slope — are included deliberately, since omitting them would make the comparison dishonest. MiniRocket is given linearly interpolated fixed-length input, which is its best case; that interpolation is precisely the cost the signature approach claims to avoid, and granting the baseline its best case is the only way to measure that cost.

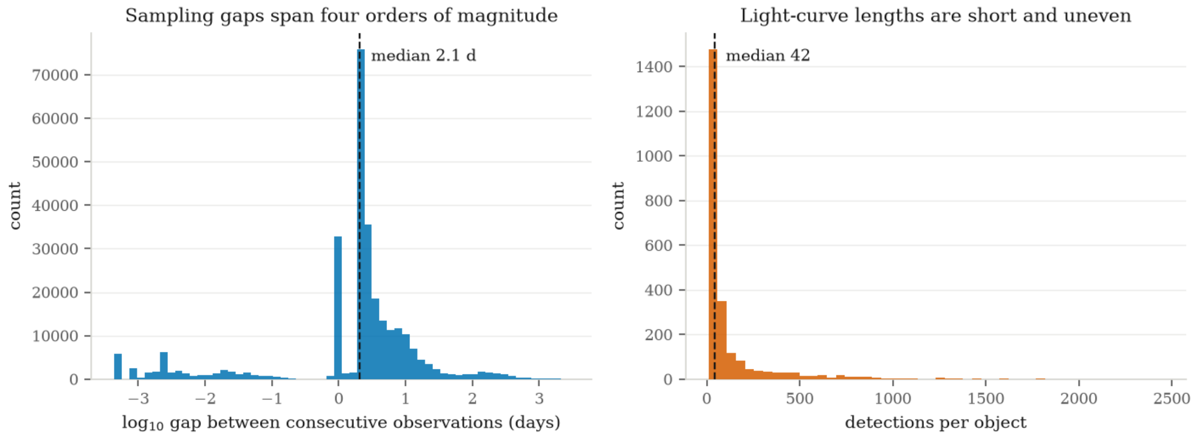

Figure 1.2: The sampling gap distribution is bimodal and spans four orders of magnitude: repeated exposures within a single night near 10^{-3} days, the main survey cadence at 1 to 10 days, and seasonal visibility windows at 100 to 1,000 days. Same-night exposures occupy almost no arc length and so barely move the signature, whereas a fixed-grid method weights them like any other sample.

1.6 Results

All figures are balanced accuracy under stratified cross-validation with gradient-boosted trees, on 2,375 objects in eight classes. Chance is 0.125.

1.6.1 Preprocessing dominated the representation

Representation	Features	Balanced accuracy
summary + signature (lead-lag, depth 4)	729	**0.578**
summary (hand-crafted)	49	0.560
signature, per-band, raw time, lead-lag, depth 4	680	0.548
MiniRocket (interpolated grid)	9,996	0.538
signature, per-band, raw time, depth 4	60	0.524
signature, per-band, unit time, depth 4	60	0.455

Table 1.2: Head-to-head comparison. The gap between the last two rows is the cost of discarding event duration by scaling time to $[0, 1]$; the gap between rows three and five is the gain from the lead-lag transform.

Scaling time to $[0, 1]$ costs 0.069, because it discards the duration of the event, which the hand-crafted baseline retains as `t_span`. The first version of this pipeline did exactly that, and would have reported a far worse number as a property of signatures rather than of the preprocessing. The lead-lag transform is worth a further 0.025, taking signatures past MiniRocket but not past the incumbent; it exposes quadratic variation, and roughly a third of the sample (AGN and CV) is stochastically variable rather than transient.

A third observation is a check on the implementation rather than a result: the *raw* and *raw time* preparations score identically to four decimal places. The signature depends only on increments, so a constant magnitude offset cannot change it. Absolute brightness can enter only through a basepoint, and introduced that way it degrades performance rather than improving it.

1.6.2 Complementarity, with a matched-noise control

Signatures alone lose to a 49-feature baseline while using fourteen times as many features. The concatenation scores above the baseline, but by 0.015 against a fold spread of 0.020, which on its own establishes nothing. The question was therefore given a dedicated test: fifty paired folds on identical splits, a Wilcoxon signed-rank test, a bootstrap interval, and a control in which the signature block is replaced by Gaussian noise of identical shape and column variance.

Feature block	Balanced accuracy
summary + signature	$\mathbf{0.5769 \pm 0.0200}$
summary	0.5623 ± 0.0167
signature alone	0.5473 ± 0.0218
summary + matched noise	0.5278 ± 0.0146

vs. summary	Difference	Folds improved
+ signature	+0.0147	38 / 50
+ matched noise	−0.0345	1 / 50

Table 1.3: Left: mean and standard deviation across fifty paired folds. Right: paired differences against the baseline, with 95% bootstrap intervals $[+0.0078, +0.0212]$ and $[-0.0396, -0.0297]$ respectively.

The control is what makes the result readable. A gradient-boosted ensemble handed 680 extra candidate splits might improve for reasons unrelated to their contents; it does not. Six hundred and eighty columns of noise *cost* 0.0345, while the same number of signature columns *gain* 0.0147, under identical folds and identical dependence structure.

One caveat belongs beside the p-value (9.5×10^{-5}): repeated cross-validation folds reuse the same 2,375 objects and are not independent, so both it and the bootstrap interval are optimistic as formal inference. The comparison that does not rest on that assumption is the one against the noise control, and it is the one the conclusion leans on.

1.6.3 What the prediction got right and wrong

The prediction registered before any data was touched was that signatures should gain most where sampling is irregular. Taken alone they do not beat the incumbent, so the strong form of the claim fails. What survives is weaker and more interesting: signatures carry information the hand-crafted features do not have. Not a better representation — a different one.

Bibliography

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